# Parameter Identification and Model Validation of Cholera Epidemic Dynamics

## Parameter Estimation, Sensitivity Analysis, and MCMC Diagnostics

Oliver NALUNKUMA

**Supervisor:** Dr. Miracle AMADI

**Course:** Model Fitting and Validation

African Institute for Mathematical Sciences (AIMS Rwanda)

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# Introduction

Cholera remains a significant global health threat, characterized by acute watery diarrhea caused by the ingestion of *Vibrio cholerae*. Unlike direct-transmission diseases, Cholera dynamics are uniquely driven by an **aquatic pathogen reservoir**.

## Objectives

In light of the 2015-2016 outbreak in Tanzania (over 24,000 cases), this study aims to:

- **Parameter Identifiability:** Use Markov Chain Monte Carlo (MCMC) and Least Squares methods to ensure the model parameters are uniquely identifiable from available data.
- **Threshold Analysis:** Determine the stability of the Disease spread to inform intervention timing.

**Assumptions**

- **Closed Population.**
- **Homogeneous Mixing** Susceptible individuals have an equal probability of contact with contaminated water.
- **Dual Excretion:** Symptomatic ($I_s$) and Asymptomatic ($I_a$) individuals contribute to the aquatic pathogen population ($B$) at rates $\alpha_1$ and $\alpha_2$.
- **Vital Dynamics:** Natural birth ($b$) and death ($\mu$) occur, maintaining population turnover.

# Mathematical Model and Parameters

**Cholera System of ODEs**

$$\frac{dS}{dt} = bN - \frac{\beta BS}{\kappa + B} - \mu S \tag{1}$$

$$\frac{dI_s}{dt} = \frac{p\beta BS}{\kappa + B} - (\mu + r_1 + d)I_s \tag{2}$$

$$\frac{dI_a}{dt} = \frac{q\beta BS}{\kappa + B} - (\mu + r_2)I_a \tag{3}$$

$$\frac{dR}{dt} = r_1 I_s + r_2 I_a - \mu R \tag{4}$$

$$\frac{dB}{dt} = \alpha_1 I_s + \alpha_2 I_a - (\delta + \phi)B \tag{5}$$

**Human Population Parameters:**

- $b, \mu$: Birth and natural death rates.
- $\beta$: Transmission rate (environment to human).
- $\kappa$: Half-saturation constant (pathogen concentration).
- $p, q$: Probabilities of $I_s$ and $I_a$ infection.
- $r_1, r_2$: Recovery rates for $I_s$ and $I_a$.
- $d$: Disease-induced mortality rate.
- $\frac{B}{\kappa+B}$: Ratio of Vibrio cholerae concentration.

**Aquatic Pathogen Parameters:**

- $\alpha_1, \alpha_2$: Excretion rates by $I_s$ and $I_a$.
- $\delta$: Rate of water treatment/intervention.
- $\phi$: Natural decay rate of *V. cholerae*.

# Sensitivity Analysis of $R_0$

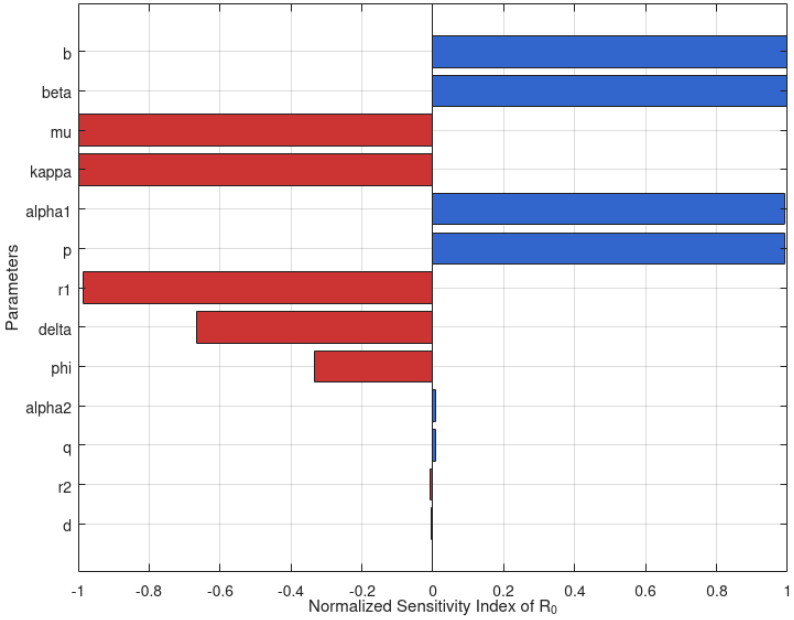


Figure 1: Normalized sensitivity indices.

**Insight:** The recruitment rate ($b$) and transmission rate ($\beta$) are identified as the primary drivers of epidemic growth.

Table 1: Sensitivity Indices for $R_0$.

| Parameter | Index | Status |
|---|---|---|
| $\beta$ | +1.0000 | Estimated |
| $b$ | +1.0000 | Estimated |
| $\alpha_1$ | +0.9421 | Estimated |
| $p$ | +0.8145 | Estimated |
| $q$ | +0.1855 | Estimated |
| $r_2$ | -0.4999 | Estimated |
| $\kappa$ | -1.0000 | Fixed |
| $\delta$ | -0.6667 | Fixed |

**Selection Criteria:**

- Parameters with $|S_i| > 0.15$ were selected for estimation.
- High indices indicate that small changes in these values significantly impact the reproduction number.

The $R_0$ true value was 5.7 and 6.5 was the estimate with the Least square.

# Model Fitting: Initial Guess

- $b$: $6.2 \times 10^{-5}$
- $\beta$: 0.30
- $\alpha_1$: 1.2
- $r_2$: 0.44
- $p$: 0.63
- $q$: 0.27

- **Proximity:** The guesses are intentionally chosen within the biological range to ensure the local optimizer does not converge to a trivial solution.
- $\sigma = 0.05$

The objective is to minimize the Residual Sum of Squares (RSS):

$$\min_{\theta} \sum_{i=1}^{n} (y_i - f(t_i, \theta))^2$$

# Model Fit

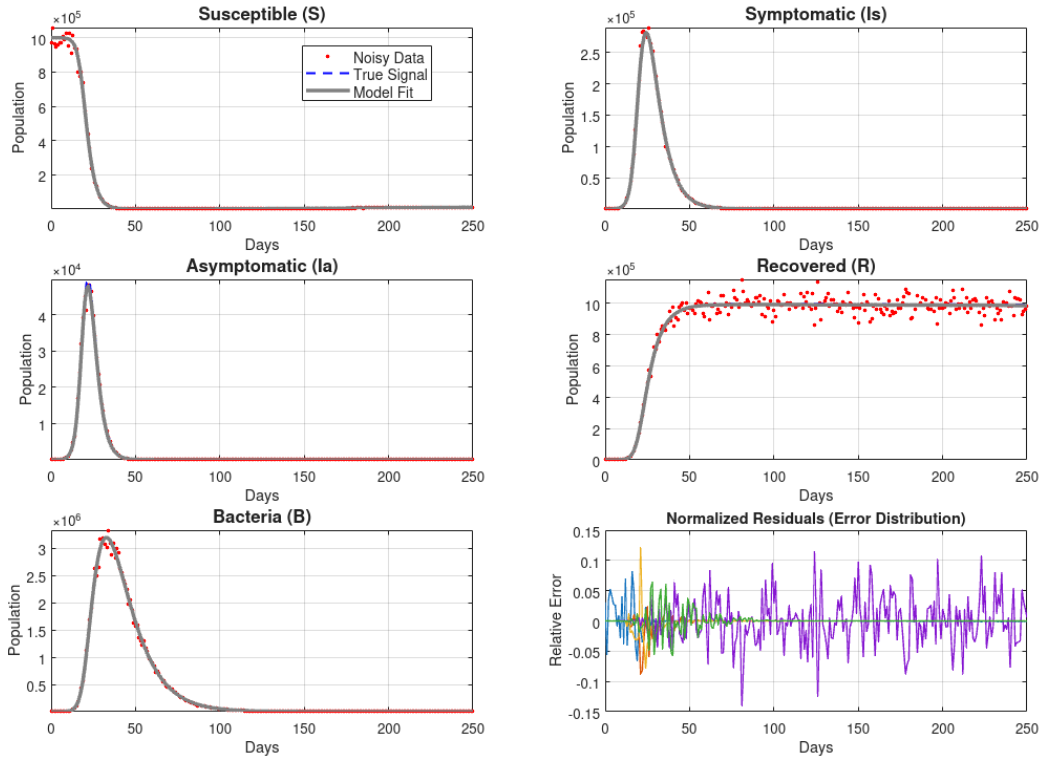


Figure 2: Least Square, True data and Noisy data trajectory comparison for $(S, I_s, I_a, R, B)$.

The Least Squares fit effectively recovers the "True" trajectory from the noisy data, especially at the Day 25 peak.

- **Residual Analysis:** The residuals (errors) show a random distribution around the zero-axis, indicating no systematic bias in the model structure.
- **Error Homogeneity:** The constant spread of residuals confirms that the model's predictive power is consistent throughout the epidemic duration.
- **Biological Consistency:** The transition from $S \to R$ is perfectly captured, validating the recovery rate $\alpha_1$.

# Parameter Estimation Results: LSQ versus MCMC

Table 2: Comparison of Cholera Epidemic Model Parameters

| Parameter | True Value | LSQ Estimate | MCMC Mean | 95% Credible Interval |
|---|---|---|---|---|
| $b$ | 0.000072 | 0.000066 | 0.000076 | $[0.000071, 0.0000724]$ |
| $\beta$ | 0.35 | 0.32 | 0.37 | $[0.34, 0.38]$ |
| $\alpha_1$ | 1.5 | 1.4 | 1.427 | $[1.39, 1.67]$ |
| $r_2$ | 0.5 | 0.46 | 0.50 | $[0.45, 1.7]$ |
| $p$ | 0.7 | 0.644 | 0.68 | $[0.62, 1.17]$ |
| $q$ | 0.3 | 0.27 | 0.3 | $[0.233, 0.33]$ |

**Note:**

The wide credible interval for $r_2$ and $p$ indicates lower practical identifiability compared to $b$ and $\beta$.

The MCMC estimate for $R_0 = 5.01$.

# MCMC diagnostics: Trace-plot Analysis

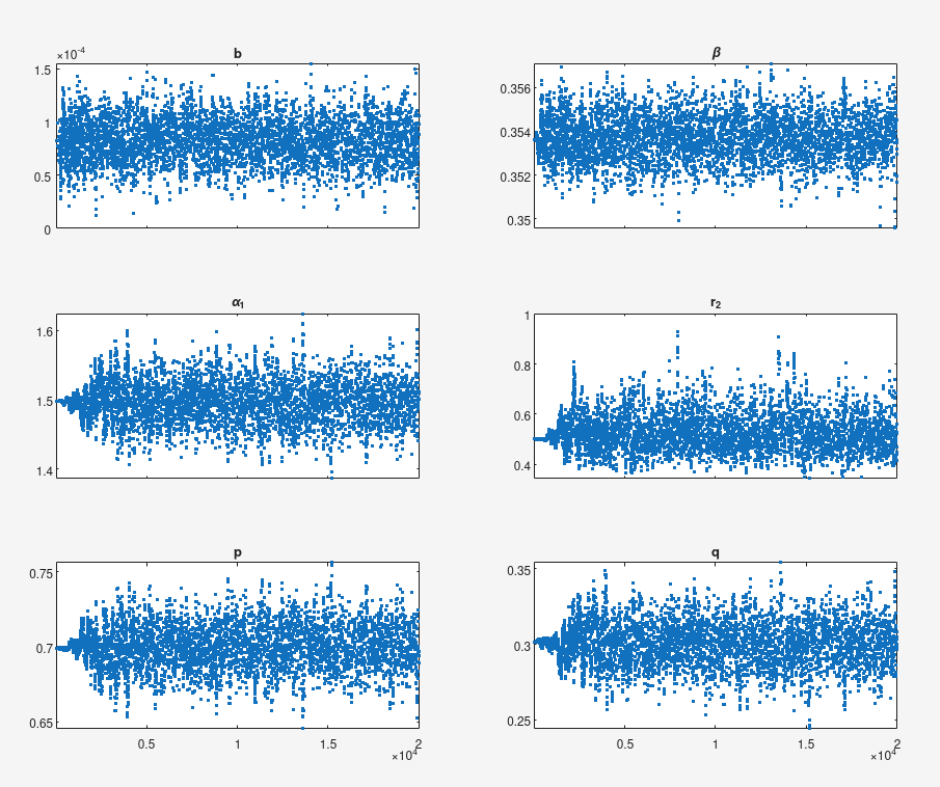


Figure 3: Parameter trace plots.

**Statistical Interpretation:**

- **Uniform Convergence:** The fact that **all six** parameters reach stationarity simultaneously proves the global stability of the MCMC algorithm.

No "drifting" or "bottlenecks" are observed in any individual parameter trace, validating the use of these samples for posterior inference.

# MCMC Diagnostics: Pairwise Interactions

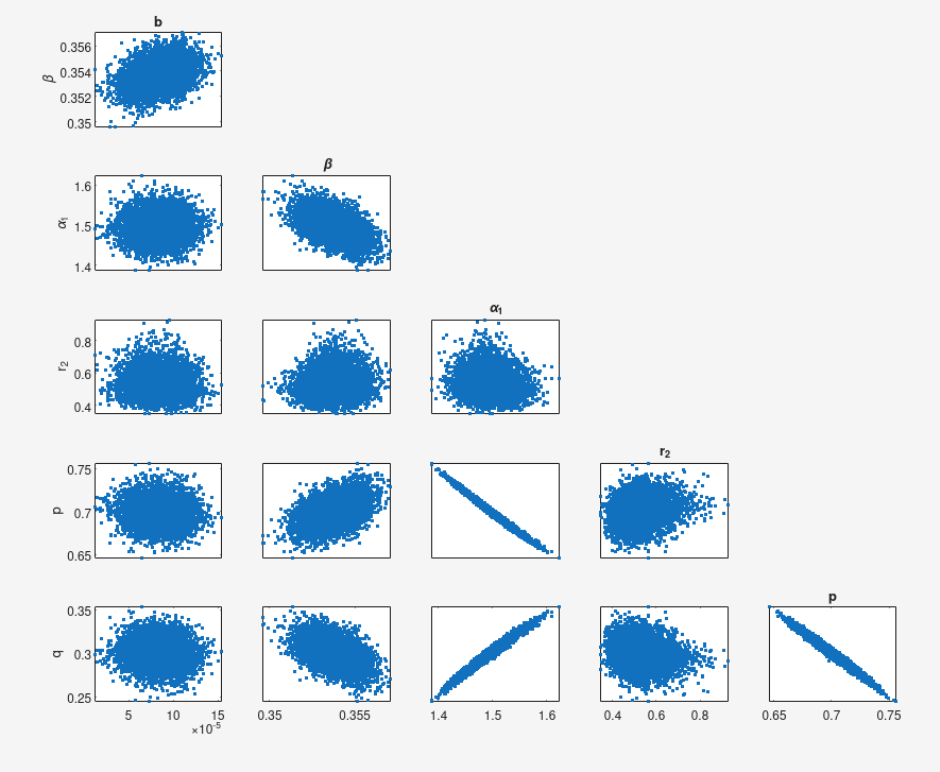


Figure 4: Pairs plot of joint posterior distributions

**Correlation and Identifiability Insights:**

- **Numerical Robustness:** The tight, localized contours suggest that the parameters are uniquely identified by the data, minimizing the risk of over parameterization.
- **Trade-off Ridge:** Any linear trends indicate a trade-off where different parameter combinations could yield similar epidemic curves, but the localized clusters show these were successfully constrained.

# MCMC Diagnostics: Autocorrelation Analysis

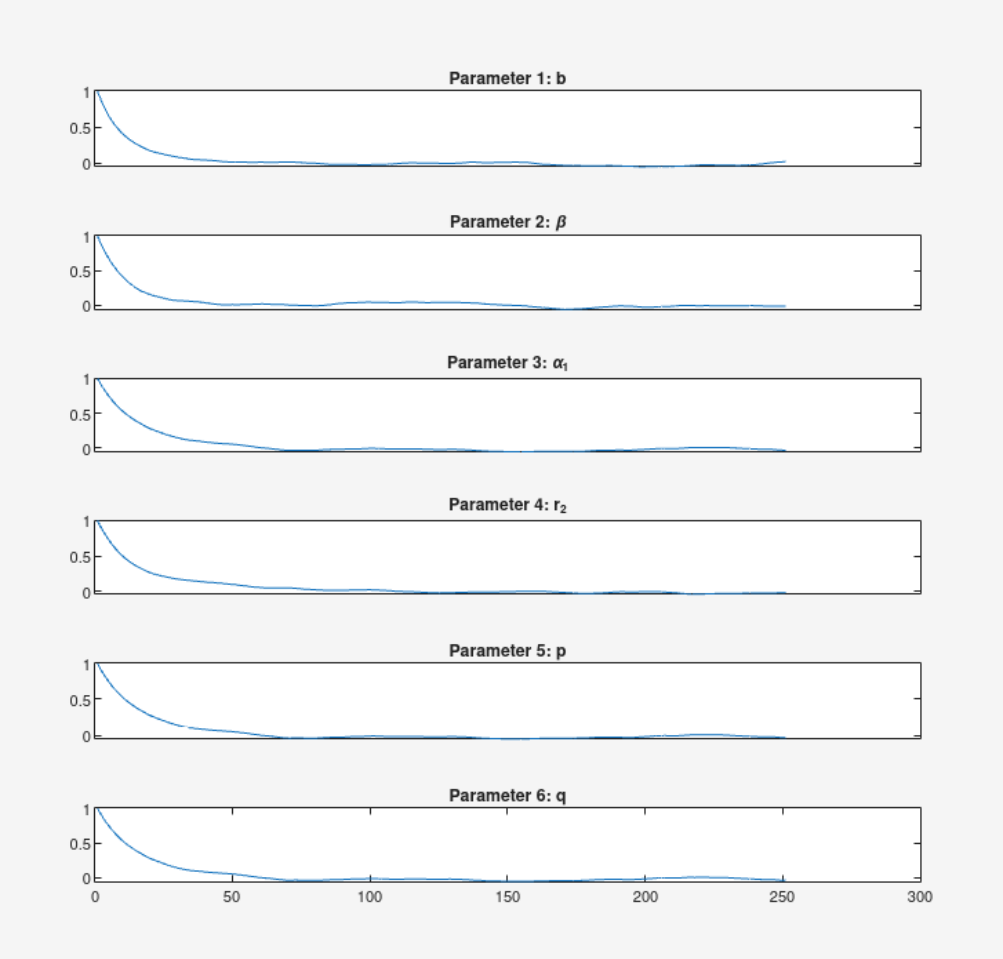


Figure 5: Autocorrelation function for 250 Lags

**Statistical Interpretation:**

- **High Sampling Efficiency:** The MCMC chain is exploring the parameter space without "sticking."
- **Algorithm Tuning:** This confirms that the proposal distribution is well-tuned to the target posterior, a direct result of the **Adaptive Metropolis** algorithm used in the study.
- **Effective Sample Size (ESS):** Low persistence in the chain increases the ESS, which minimizes the variance of our posterior estimates and ensures the means are reliable.

# MCMC Diagnostics: Marginal Posterior Distributions

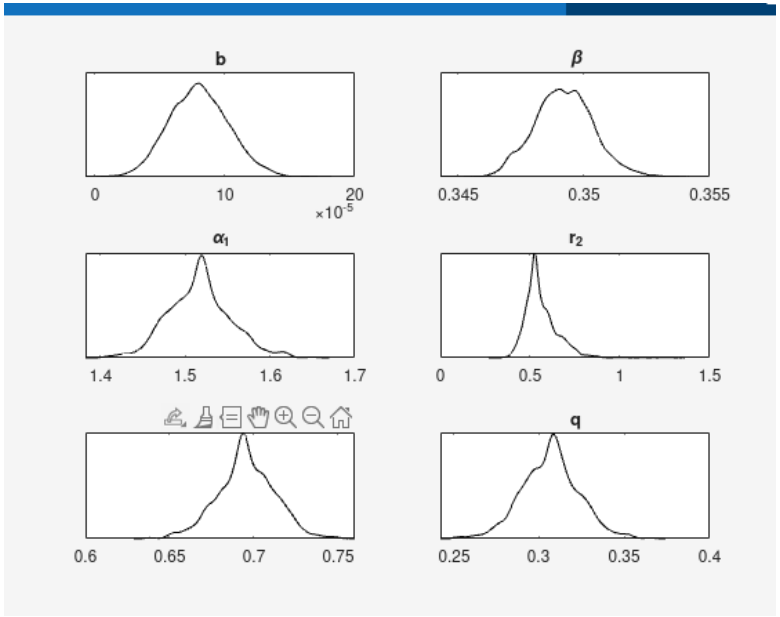


Figure 6: Posterior distribution of unknown parameters with 10000 iterations.

- **Parameter Identification:** The "bell-shaped" curves represent the most likely values for our unknowns parameters. The peak of each curve is the **Posterior Mean**.
- **Uncertainty Quantification:** The width of these distributions indicates our certainty. A narrow "Normal" curve suggests the data strongly identified that specific parameter.
- **Propagation through the Model:** These distributions are what we "propagate" to get the gray uncertainty bands seen in the epidemic trajectories.
- **Consistency:** The smooth, unimodal (one-peak) shape confirms that the MCMC chain converged to a stable solution for the Tanzania outbreak data.

# MCMC Diagnostics: Posterior Predictive Analysis

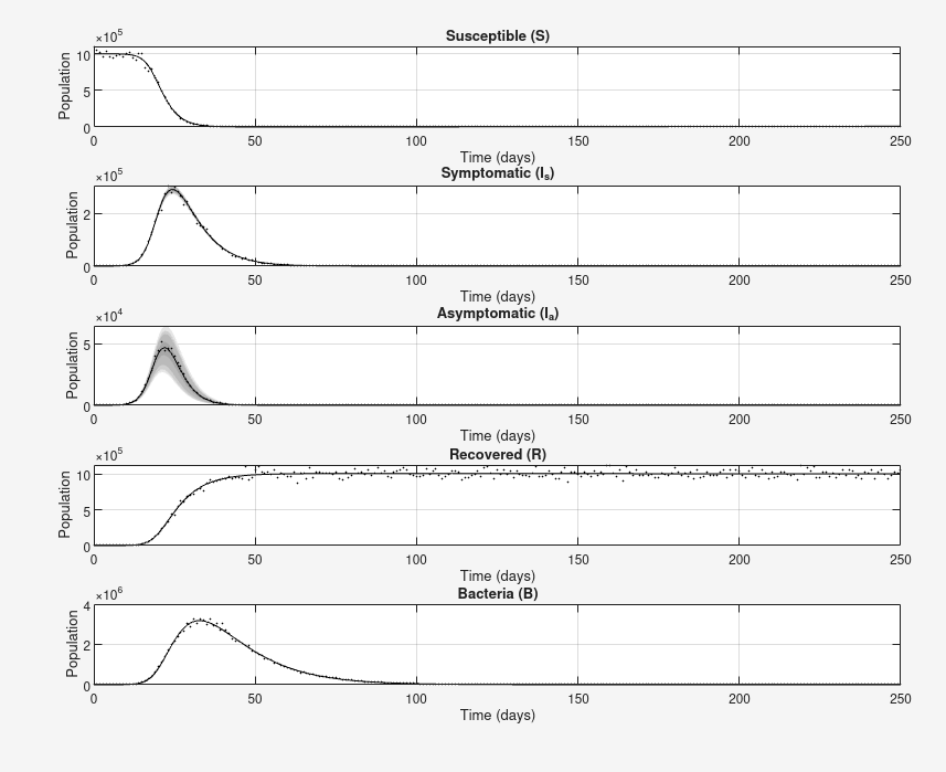


Figure 7: Posterior predictive trajectories with 95% credible intervals.

**Statistical Interpretation:**

- **Sample Propagation:** We take the thousands of unbiased parameter sets $(b, \beta, \alpha, p, q, r_2)$ from the MCMC chain and run them through the ODE system.
- **Trajectory Estimation:** This gives us the predictive distributions for the state variables:
- **Uncertainty Quantification:** The "gray clouds" (95% credible intervals) represent our structural and parametric uncertainty.

# Public Health Insights & Conclusion

## Theoretical Conclusion:

- The model confirms that $R_0 < 1$ leads to a stable disease-free state, whereas $R_0 > 1$ ensures disease persistence (endemicity).
- MCMC diagnostics prove the parameters are uniquely identifiable, making the model a reliable tool for public health forecasting.

## Public Health Policy Recommendations:

- **Target Asymptomatic Carriers:** Since $I_a$ individuals contribute to the pathogen reservoir silently, sanitation and screening are vital for long-term control.
- **Rapid Response:** Intervening before the Day 25 peak is essential to reduce the total number of infections.

**Thank you for your attention!**

# References

## Primary Study


- **Marwa, Y. M., Mwalili, S., & Mbalawata, I. S. (2018).** Markov Chain Monte Carlo Analysis of Cholera Epidemic. *Journal of Mathematical and Computational Science*, 8(5), 584-610.


## Supporting Literature


- **Laine, M. (2008).** *Adaptive MCMC Methods with Applications in Differential Equations.* Helsinki University of Technology (Source for MCMC diagnostics).
- **Shuai, Z., & van den Driessche, P. (2012).** Global stability of infectious disease models using Lyapunov functions. *SIAM Journal on Applied Mathematics*, 72(3), 675-692.
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- **Codeco, C. T. (2001).** Endemic and epidemic dynamics of cholera: the role of the aquatic reservoir. *BMC Infectious Diseases*, 1(1), 1-14.